up by AI as we voyage to the Moon and the planets, and explore possibilities in Space. AI is also being used in NASA's next rover mission to planet Mars.

National Geographic Society and Microsoft are partnering to explore how AI can help us understand, engage and protect Earth.

AI is helping the oil and gas industry to preserve the ecosystem while discovering new resources.

Robotics and AI have been replacing traditional research and exploration methods in ocean science and technology.

Education. AI is being used to improve education systems. Traditional techniques might be replaced by personalised, adaptive learning to tailor individual students' strengths and weaknesses. Machine learning can be used to identify students and focus extra resources on weaker ones.

AI-based robo-readers are being used in essay grading in schools. The approach involves pairing human intelligence with AI to improve the overall grading system and help students accomplish more.

Automotive. ANI-assisted automotive technology is being employed in driverless cars. Recently, IBM developed an IoT for automotive—a program to eliminate driver errors through connectivity. Since many accidents are caused by human errors, researchers are trying to find ways to minimise human errors using AI algorithms.

In the future, your car is likely to have well-packaged complex computer programs. An automated vehicle uses AI, sensors and global positioning system coordinates to drive itself without a human operator.

Video games. You might already know about Deep Blue, IBM's chess-playing supercomputer, which beat international grandmaster Garry Kasparov in the late 1990s.

Chinook, a program developed at University of Alberta, Canada, can beat any human player at the game of checkers. There is also a computer program called Maven for scrabble game. These are perfect examples of AI.

More recently, AlphaZero, an ANI developed by DeepMind, won 100 games in a row against the world's current best chess program.

Almost every modern video game has an AI component. Video game reviews are based on the quality of AI.

Healthcare. A new type of AI algorithm is being used by Google's Medical Brain to make predictions about the likelihood of death among hospital patients. This technology is the latest attempt to revolutionise healthcare. AI programs are being developed and used in diagnoses, treatments, drug development, and patient monitoring and care.

Today, most doctors use ANI programs. These assist doctors in accurately diagnosing cancer and various other diseases. Then, there is a robot chemist that uses machine learning to study new molecules and reactions.

Military. AI and robotics are enabling new military capabilities and strategies including intelligence, surveillance and even nuclear weapon systems. AI is used in autonomous weapons and sensing systems.

There is a lot of military AI research and development going on around the world to keep fast-paced advances in machine learning.

India and AI

As per a report by Accenture, AI holds the potential to add US$ 957 billion or 15 per cent of India's current gross value by 2035.

Union Budget 2018 announcements include national programmes to conduct research and development on technologies like machine learning, AI and others.

NITI Aayog, the nation's think-tank and premier policy-making body, is working on new technologies for the development of the economy.

The government is working on AI initiatives to put India on the global map with regards to AI, and to promote it in health, education and agricultural sectors.

India-made Miko by Emotix, a companion robot, is an example of technologies that incorporate AI. Miko engages, educates and entertains children besides talking to and playing games with them. It is equipped with answers to basic questions related to general knowledge and academics.

There is a humanoid robot called Rashmi, developed recently in India. It is the first Hindi-speaking robot who is hosting a show on Red FM since December 2018.

The way forward

AI is becoming a disruptive force that is redefining modern industry. Importance of AI technology is being felt across a broad spectrum of industries. From voice-powered personal assistants like Alexa to technologies such as behavioural algorithms, suggestive Internet search algorithms and autonomous vehicles, there is a lot of scope for applications of AI today. Robots built to look and act like humans are getting a lot of attention, and making splashy headlines and appearances.

As far as India is concerned, initiatives by the government and its roadmap for national AI programmes along with private players are expected to bring a revolution to the Indian industry.

AI is still in the developing stages. The market is not easily quantifiable and yet there are plenty of opportunities available for AI. There is hope that AI applications will keep serving humans in the most beneficial way going forwards. EFY

Miko, a companion robot (www.amazon.in)

Some leading developers of AI

- Acorn
- Apple
- Argo AI
- Cloudwalk
- DJI
- Emotix
- Google
- Hanson Robotics
- IBM
- Kreditech
- Microsoft
- SenseTime
- Soundhound
- Tesla
- Toshiba
- Ubtech Robotics